4.4 Forever loop

In this lesson

You will learn:

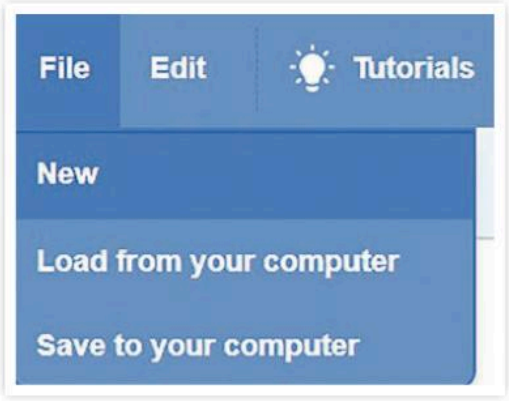
- how to change the program so the sprite moves 'forever'.

Last lesson you wrote a program. The program started when you clicked on the sprite. The sprite moved to a new place. The sprite made a sound. These things happened one time.

Now you will make a new program. In this program the sprite will move over and over again. The action will repeat.

New program

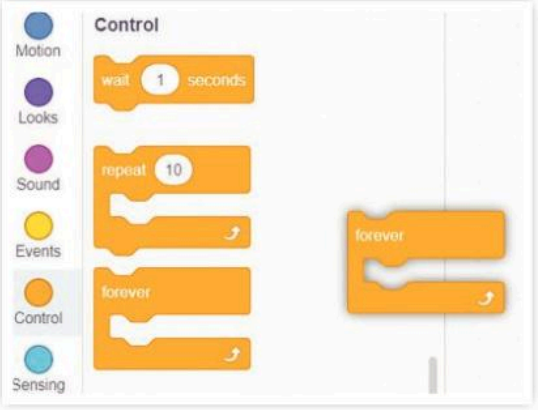
Open the file menu. Click on 'New' to make a new program.



Control blocks

Click on the orange button. You will see the orange 'Control' blocks. Find the block that says 'forever'. This block is called the **forever loop**. Drag it to the middle of the screen.

A forever loop is a way to control the program. Commands inside the forever loop will repeat.



Add commands

Here are the commands you want to repeat:

- Make the sound 'pop'.
- Go to a random position.

Put these blocks inside the forever loop. They are the same blocks you used in the last program.

Wait

Add one more block inside the forever loop. You will find it with the orange control blocks. It says 'wait 1 seconds'. This will make the sprite wait for 1 second between jumps.

The completed program looks like this.





Activity

Start a new program. Choose a sprite and a background.

Make the program shown on this page. Run the program to see what happens.

Think again

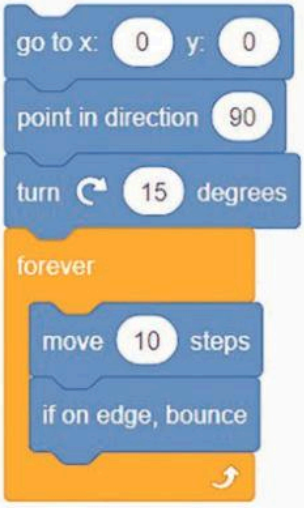
Say what happens when you put blocks inside a forever loop.



Extra challenge

Here is a new program that uses the forever loop.

- Make this program.
- Add a start 'Event' to the program.
- Change the numbers in the program and see what happens.



4.5 Plan and make

In this lesson

You will learn:

➔ how to make a new program to match a plan.

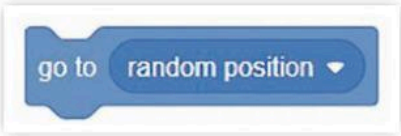
Plan

Here is the plan.

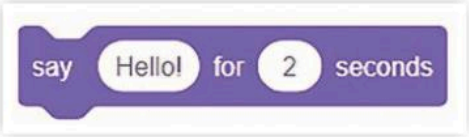
- The sprite will jump to a random place.
- The sprite will say "Hello!"
- The sprite will keep moving towards the mouse pointer.

First commands

The sprite will start in a **random** place. Random means you do not know where it will go.



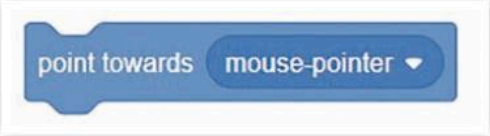
The sprite will say "Hello!"



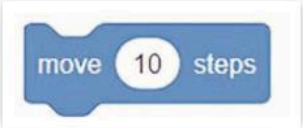
Find these blocks and fit them together.

Repeated commands

Here is the block that makes the sprite point towards the mouse pointer.



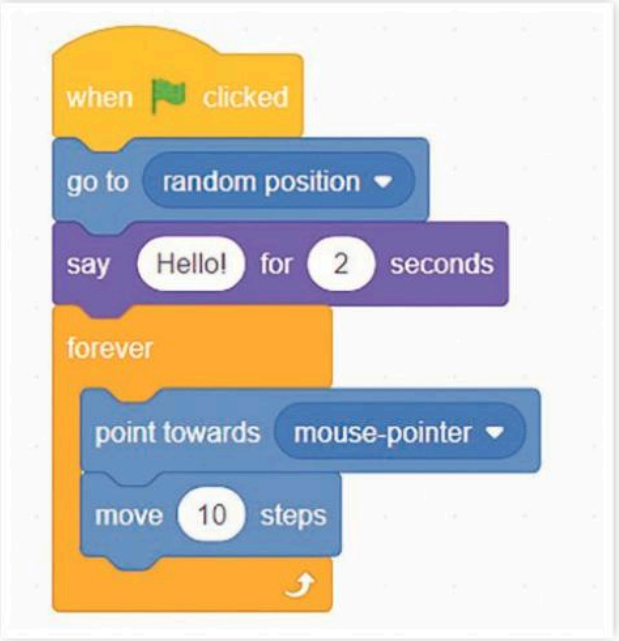
Here is the block to move 10 steps.



These two blocks will go inside a forever loop.

Completed program

Here is the program with all the pieces in place. It has a start 'Event' – the green flag.





Activity

Start a new program. Choose a sprite that looks like a fish. Choose a background that looks like underwater.

Make the program shown on this page. The program will make the fish swim about underwater.

Think again

Think of one change you could make to the program you made. What would happen if you made this change?



Extra challenge

Add a jellyfish sprite to the underwater scene. Make a program for this sprite.

- Use the same blocks as before.
- Take out the block that says 'follow the mouse pointer'.
- Add a block that says 'if on edge bounce'.
- Make the number of steps 2 instead of 10.

